

Section A. Question 1 Answers (10 Marks)

Based on the table, answer the following questions:

(a) "Nearly all amino acids have an asymmetric carbon atom." What is (are) the consequence(s) of this observation? [4]

- α carbon is the asymmetric carbon atom;
- bonded to four different substituent groups;
- thus a chiral center;
- compounds with chiral center occur in two different isomeric forms;
- non-superimposable mirror images of each other;
- Identical in all chemical and physical properties;
- except for the direction in which plane polarized light is rotated / optically active;
- levorotatory and dextrorotatory;

(b) The amino acids of a polypeptide affect the shape of the protein. Predict where each of the following amino acid pairs will be found in the protein and explain why.

(i) One amino acid is arginine and the other is serine. [1]

- facing toward the outside;
- arginine is a charged amino acid, serine is polar uncharged;

(ii) One amino acid is alanine and the other is isoleucine. [1]

- folded toward the interior;
- both are non-polar/hydrophobic;

(c) "Most proteins incorporate small quantities of glycine." Explain why. [2]

- Glycine has the smallest side chain of any amino acid;;
- Is critical in allowing polypeptide chains to make tight turns;
- Or to approach one another easily;

(d) Sickle cell anemia results from a single base substitution. The resultant β globin chain, of haemoglobin S, has a valine residue instead of a glutamic acid. What do you expect the effect on the β globin chain might be if the mutation were to involve

(i) leucine, instead of valine? [1]

- leucine similar in structure to valine;
- protein to have similar abnormal properties;

(ii) aspartic acid, instead of valine? [1]

- aspartic acid is similar to glutamic acid;
- very little effect might be expected

Question 2

- a) Explain which enzyme, **A** or **B**, has the greater affinity for its own substrate? [3]
- The **K_m for A is 0.3 umoles** of glucose,
 - while the **K_m for B** would be **2.0 umoles**;
 - therefore it takes significantly more glucose to reach the same one-half V_{max} for enzyme B;
 - **enzyme A would have the greater affinity** for the substrate glucose
 - **Greater chance of forming ES complexes**;
 - Hence **greater chance of product formation** ;
 - Because ES complex formation is the **rate determining step**;
- b) If enzyme **A** were subjected to non-competitive inhibition by the drug insulineride, explain what you would expect the V_{max} to be? [2]
- The inhibitor would bind to a site other than the active site of the substrate,
 - and remove a fixed amount of active enzyme from the mix
 - inhibition cannot be overcome by increasing substrate concentration
 - and therefore the **V_{max} would be less** in the presence of insulineride.
- c) From the plot shown by enzyme **B**, explain the advantage of having enzyme **B** in a cell. [2]
- The plot shown by enzyme B is a sigmoid curve which shows that enzyme B is an allosteric enzyme;
 - Its activity is therefore modified by the (noncovalent) binding of an allosteric effector (activator or inhibitor) at a site other than the active site;
 - This binding mediates conformational changes, altering its catalytic or binding properties;
 - Thus enabling enzyme B to be a key regulator of metabolic pathways in the cell;
 - and its catalytic properties can be adjusted to meet the immediate needs of a cell.
- d) What is a cofactor and explain the role of magnesium as a co-factor in enzyme **A** activity. [3]
- Cofactor is a small molecule that determines the catalytic activity of many enzymes;
 - can be subdivided into two groups: metals and small organic molecules;
 - magnesium is a metal
 - which can act as an effective electrophile;
 - its +ve charge helps in binding to ATP which is polar/charged;
 - Thus help to orientate the substrate molecule within the active site of the enzyme;
 - And polarizes the substrate to promote catalysis;

Question 3

- (a) Explain what is meant by the term “proteome”. [1]

It means the **complete collection of proteins** [1/2] encoded by the **genome of an organism / cell**. [1/2]

- (b) Outline the steps carried out to obtain the proteome of the α T3 and L β T2 mouse pituitary cells. [4]

Step 1) **Culture the α T3 and L β T2 mouse pituitary cells** under suitable conditions / any ref in obtaining a large no. of cells so as to obtain sufficient proteins [1/2]

Step 2) **Extract** and solubilize **proteins** from the respective cells [1/2]

Ref to any known method: homogenizers/blenders, osmosis-based method and sonication. [1/2]

Under **conditions** that **prevent proteolysis** of the proteins [1/2]

Ref to any known example: Lyse / homogenize cells under very low temperature / snap frozen and ground under liquid nitrogen / protease inhibitors to prevent enzymatic degradation of proteins [1/2]

Step 3) **Isoelectric focusing** to separate proteins according to their isoelectric points [1/2] using **immobilized pH gradient strips** [1/2]

Step 4) Run **SDS-PAGE** [1/2]

Step 5) **Stain** proteins [1/2]

Ref to any known staining agent: **Coomassie Blue stain / SYPRO Ruby / silver stain** (note: silver stain - incompatible with mass spectrometry) [1/2]

Imaging by computer software or by visually [1/2]

- c) With reference to figure 1 and 2, explain what can be done to improve the resolution of the gel and subsequent identification of nuclear proteins [1]

Ref to **low abundance** of **nuclear proteins** [1/2]

Remove high abundance undesirable proteins (e.g. actin) [1/2] which may mask the desired nuclear proteins by **column chromatography** (e.g. affinity – or ion exchange chromatography) [1/2]

Pre-fractionation of cells by centrifugation and analyze proteins from the **nucleus** only [1/2]

Reject: use of narrow pH gradient range IPG strips during IEF or column chromatography to extract nuclear proteins

- d) On figure 4, circle one possible protein spot on the that may be a nuclear protein [1]
- ei) On the diagram below, indicate the result of SDS-PAGE in lane 2. [1]
- a **single band** [1/2] **correctly positioned** wrt molecular weight [1/2]

(ii) Describe one method that may help you identify the correct protein band in lane 2. [2]

- **Western blot / immunoblot** [1/2]
- **Transfer the proteins** on lane 2 to **nitrocellulose / membrane** [1/2]
- Add **generic protein** (e.g. milk proteins or bovine serum albumin) [1/2] to prevent **non-specific binding between membrane and antibodies** [1/2]
- The antibody has an **enzyme** [1/2] (e.g. alkaline phosphatase or horseradish peroxidase) [1/2]
- or **fluorescent dye** attached to it
- **Wash away excess antibodies** [1/2]
- Location of the antibody (and hence the desired protein) is revealed by **incubating it with a substrate that the attached enzyme** [1/2] converts to a **colored product** (that can be seen and photographed) [1/2]
- or **excitation** and the **subsequent emission of the fluorescent probe**

Question 4

(a) Suggest how Oseltamivir works to halt the proliferation of the H1N1 virus. [1]

- Antiviral drugs like *Oseltamivir* are potent competitive inhibitors of Neuraminidase, particularly when they are used in the first 48 hours after illness onset, e.g., before the virus is able to release itself from the initial host cells and infect other cells.

IDEA: In the absence of the viral Neuraminidase, the new formed viruses would remain linked to the sialic acids residues in the cell surface or in the glycoproteins in the respiratory tract mucus, and they would get trapped and impeded of infecting other cells.

(b) In the development of anti-viral drugs, it is often important to employ high throughput methods to determine binding partners to a query protein. State one such method. [1]

- Yeast-2-hybrid

(c) Explain how RNA interference can be used to characterize the cellular roles of proteins. [3]

- Introduce small double-stranded RNA; corresponding to part of the protein mRNA sequence;
- so as to suppress protein translation
- see phenotype change to deduce possible function of protein

(d) Explain why RNA interference is not the method of choice to elucidate the cellular roles of membrane proteins on the viral envelope. [1]

- Virus have a high rate of mutation
- Viruses are obligate pathogens and some may intergrate into the host genome like HIV to form provirus for an extended period of time.

(e) Discuss four characteristics of an efficient high throughput cloning system. [4]

Fidelity

- From the standpoint of the end-users, the defining requirement for a high-throughput cloning system is a high fidelity of transfer of cloned DNA from master clones to expression plasmids. Moreover, if creating a collection of expression clones requires thousands of transfers, the transfer efficiency of cloned DNA must approach or equal 100%

Ease of Use

- The ideal cloning system should be simple to use and involve a minimal number of steps. Pilot studies and clone transfers of small groups of ORFs should be achievable by a junior-level scientist using a multichannel pipette. In creating very large ORF clone collections that include many thousands of genes, however, the use of automation to assist the cloning process is inescapable. Removing human handling of samples from the cloning process helps to create a more systematic process, results in a process with fewer errors, which integrates better with a clonetracking database.

Reliability

- If the cloning system is available as a kit, the components of the kit must be stable and reliable. It is especially desirable if the reagents can be purchased in large quality-controlled lots, so that once optimal conditions are established, they do not have to be readjusted in the middle of a production run.

Validation

- Sequence validation of master clones is arguably the most expensive stage of the cloning process. The goal of sequencing is to determine which clones are acceptable for use in protein expression experiments. Even small changes in the amino acid sequence of a protein can have profound effects on its biochemical activity and function.

Flexibility

- A powerful advantage of recombinational cloning systems is their ability to support the transfer of genes into virtually any type of expression vector, supporting the broadest possible range of experimentation. For some cloning systems, a varied collection of expression vectors have already accumulated. Nevertheless, the vectors available may not be suitable for novel experiments, so it is especially desirable for users to be able to easily create their own expression vectors and adapt them for use in the cloning system.

Minimizing Undesirable Properties of Cloning Systems

- In many cases, particularly in high-throughput experiments, it is desirable to add polypeptide tags to one or both ends of the expressed proteins that can be used as purification tags (His6, GST, CBP, etc.), marker proteins (*LacZ*, GFP, luciferase, etc.) or as epitopes (HA, Myc, Flag, etc.). The use of fusion tags is, however, somewhat complicated by the contribution of the recombination sequences to the tags, and how stop codons are handled. It is thus important that the design of the master clones is informed by the user's needs with respect to the tags used in any expression clones that will be created. For example, if carboxy-terminal tags are desired, then the natural stop codon must be omitted, or there will not be read-through into the tag. However, the removal of stop codons for some cloning systems will prohibit the production of native protein, because all expressed proteins will have amino acids corresponding to the recombination site fused to the end of the protein. As the inclusion or exclusion of the stop codon results in different versions of the 3' PCR primer used in amplifying the ORF, this decision must be made at the very beginning of the project.

Cost

- The importance of cost should not be underestimated. Although a reaction cost of a few dollars per sample is rarely a problem for a handful of reactions, when many thousands of samples are involved, truly daunting costs can accrue. This is especially true for the reaction costs involved in transferring ORFs from one vector to another, which will be incurred each time a new expression collection is created.

Question 5

(a) *What do you understand by the term domain?*[2]

-is a substructure that is produced by any part of the polypeptide chain that can fold independently into a compact, stable structure;

- usually has a specific function;

(b) *Explain the results shown in Figure 1.1 with regard to the binding of PI 3-kinase to phosphotyrosines in the activated PDGF receptor.* [3]

- PI 3-kinase interacts/binds to the phosphorylated tyrosines at 708 and 719;;
- Since no PI 3-kinase activity with pentapeptides with phosphotyrosines;;
- No bound PI 3-kinase/ PI 3-kinase binds to these 2 types of pentapeptides;;

Explanation:

- Under normal conditions, the binding of the PI 3-kinase to the phosphorylated tyrosine on the receptor activates the enzyme;

- We would expect this PI 3-kinase to be activated when it binds to the phosphotyrosines of the pentapeptides;
- So if there is an excess of the phosphorylated pentapeptides, then the chance of PI 3-kinase binding to them is high i.e. the chance of the binding to the phosphorylated receptor is low;
- This means that in the immunoprecipitation, low or no PI 3-kinase will be collected and hence there will be no phosphorylation of its substrate as seen in Figure 1.1;

(c) *What are the common features of peptide segments that form the binding sites for PI 3-kinase? [2]*

- Presence of a **phosphotyrosine**;; (if student mentioned tyrosine - ½m)
- Presence of a methionine (M);
- located 3 positions in the C terminal;

(d) *Name the type of protein-protein interaction between PI 3-kinase with the PDGF receptor. Explain your answer. [2]*

- surface–string interaction;;
- recognition of sequences by SH2;; / explanation that the PI 3-kinase is attached to a “string” of amino acids;;

(e) *State another type of protein-protein interaction. [1]*

- Surface-surface Or helix-helix

Section B

Question 6

(a) High concentrations of urea will unfold proteins. Explain why. [2]

- Urea functions both as an efficient hydrogen bond donor (through its -NH_2 groups);
- And as an efficient hydrogen bond acceptor (through its -C=O group);
- Disrupt hydrogen bonds that stabilize protein molecules;
- Exposes the non-polar side chains;
- Hydrophobic forces diminish;

(b) Consider the following protein sequence: (Reference can be made to the list of amino acids of Question 1 in Section A)

Leu-Lys-Val-Asp-Ile-Ser-Leu-Arg-Leu-Lys-Ile-Arg-Phe-Glu

(i) What is the arrangement of amino acids in this sequence when incorporated into a β sheet? [2]

- Amino acid sequence consists of alternating;
- non-polar and charged or polar amino acids;
- resulting strand in a β sheet would therefore be polar on one side;
- and hydrophobic on the other;

(ii) Predict how the β sheet, with the incorporated sequence of amino acids, may be arranged in a protein. [1]

- Would be positioned so that the hydrophobic side would face the protein's interior;
- Polar side would be on its surface, exposed to the water surface;

(c) Consider a polypeptide composed entirely of glycine, i.e. polyglycine. Which environment, i.e. a hydrophobic or an aqueous environment, will be more likely to promote α helix formation in this polypeptide? Why? [3]

- α helix conformation involves intrachain hydrogen bonds;
- between all the peptide bond N-H and C=O groups;
- in aqueous environment, hydrogen bonds form either internally (i.e. α helix conformation) or with water molecules;
- polypeptide assumes extended conformation;
- hydrophobic environment more likely to promote α helix formation;
- does not provide any competition to hydrogen bonding;

(d) A scientist has determined the sequence of an unknown protein. Using your knowledge of Bioinformatics, how will you predict the protein structure? [2]

- Look for a homologous protein;
- With a known 3D structure;
- in the protein data bank;
- via the Protein-Protein BLAST (blastp);

Question 7

(a) Describe a biological database. [3]

- A large organised body of persistent data;
- Associated with computerized software;
- Designed to update, query and retrieve components of the data stored within the system.
- Example of data: DNA sequences/ protein sequences/ macromolecular structures/ genome sequences/ whole genome data.

(b) Describe an example of a biological database that can be used to deduce a possible function of the novel protein and one kind of information that can be obtained from it. [3]

- Prosite;
- Provides high level of annotation/ minimum level of redundancy/ high level of integration with other databases;
- Contains information on characteristics of protein families/ domains;
- Sequence alignment of homologous regions in search of conserved motif;
- Motifs are related to biological functions such as active sites/ binding sites.
- Similar 3D structures, functions shares significant sequence similarity;

(c) Describe briefly how the protein was found to contain the domain. [2]

- Use BLAST;
- Local alignment software against amino acid sequences of proteins of known functions;
- Select for queries with significant alignments;

(d) The microorganism was later found to carry a genome that is smaller than *E.coli*. Surprisingly it has almost the same number of functional proteins. Suggest a reason for this. [2]

- The microorganism may be eukaryotic in nature;
- Genes consist of introns and exons;
- Alternative splicing;
- Different combination of exons.

Question 8

- (a) Like other forms of posttranslational modification, the covalent attachment of phosphate groups to protein targets at any of a number of surface Asp, His, Ser, Thr, or Tyr residues can profoundly affect protein behavior. State where such phosphorylation is often needed. [3]
- Phosphorylation of components involved in DNA replication and translation. Involved in cell cycle regulation
 - Phosphorylation of components involved in signal transduction/cell signaling pathway;
 - Important for protein-protein interaction via "recognition domains" Phosphorylation of the cytosolic components of NADPH oxidase, a large membrane-bound, multi-protein enzyme present in phagocytic cells plays an important role in the regulation of protein-protein interactions in the enzyme
 - Mediates inhibition protein function eg enzyme inhibition. Phosphorylation of the enzyme GSK-3 by AKT (Protein kinase B) as part of the insulin signaling pathway.
- (b) In bacteria, secreted and extracellularly oriented membrane proteins are often stabilized by disulfide bonds, i.e., covalent links between the sulfhydryl groups of Cys residues in the same or different polypeptide chains. Suggest a role for such bonding. [1]
- These can serve two roles. First, they can stabilize proteins by entropic destabilization of the unfolded conformation/ thermostability of the 3D conformation
 - Second, they serve to limit damage to a protein resulting from oxidative or proteolytic agents, thereby enhancing protein lifetime. Accordingly, disulfide bonds are routinely employed by secretory and plasma membrane proteins in numerous organisms.
- (c) The protein spot from a SDS-Page gel was cut and trypsin digested for identification using mass spectrometry. However, the protein databases were unable to identify the protein accurately. Explain how the method of Liquid chromatography coupled to mass spectrometry may determine the amino acid sequence of the unknown protein. [2]
- Cleave **peptide bond** from the **N or C-terminus of the protein** [1/2]
 - The different peptide fragments are separated by **chromatography** (e.g. HPLC) to generate a **ladder of fragments** [1/2], each differing by a **single amino acid** [1/2]
 - The peptide fragments are introduced to the mass spectrometer where the **difference in mass between the peptide fragments** corresponds to the amino acid sequence [1/2]
 - Ref to **electrospray ionization** to generate fine spray of charged droplets during the ionization of the analyte stage [1/2]

(d) Explain how Edman degradation can also be used to determine the amino acid sequence of a polypeptide [2]

- Amino acids are **labeled** [1/2] and **cleaved off from the N-terminal end** of the peptide chain [1/2]. **Individual amino acids** are then **identified** [1/2] using **chromatography** / electrophoresis [1/2] after which the sequence can be determined
- Ref to effective method to sequence proteins **less than 50 amino acids long** [1/2]

Question 9

(a) Describe how the yeast 2-hybrid is used to study protein-protein interactions in the above example. [4]

- Assembly of an active TF from 2 fusion proteins;;
- Description of bait;;
- Description of prey;;
- Detection of this assembly by activation of a reporter gene;;

(b) Explain why a variety of cDNA library is used in this study. [2]

- cDNA contain only the expressed sequences /sequences that are used for the translation of functional protein;;
- variety means different cDNA coding for different proteins are used;;

(c) Explain how the variola G1R can cause death in humans. [2]

- Variola binds to human nuclear factor, preventing the activation of inflammatory responses;
- Fn of transcription factor is to activate transcription of genes that are involved in defence/ killing the viruses;
- Cells in humans cannot kill off the viruses;
- Viruses multiply and cause cell damage; /≡

(d) Explain what it meant by the G1R protein being highly conserved and the significance of this conservation to pathogenic orthopoxviruses. [2]

- Highly conserved – the base sequence of the G1R proteins is highly similar in the different pathogenic orthopoxviruses;;
- This protein allows these viruses to be highly successful in invading the host cells and allow the viruses to propagate and hence they are selected for;;

Section C (20 marks)

Question 10

- (a) Describe how bioinformatics can be used to analyse a novel protein. [10]
- Bioinformatics is the analysis of biological information utilising computer softwares and databases;
 - Involves the comparison of the amino acid sequence of the novel protein against a database of known proteins;
 - BLAST; local alignment of the subject sequence against query;
 - In search of homology;
 - Indicated by lowest E-values and highest S-scores;
 - Provides information on structural/ chemical/ physical properties such as molecular weight/ isoelectric point;
 - Protein motif -definition
 - Protein domain -definition
 - Protein family- definition
 - Post-translational modification;
 - 3D structure;
 - Multiple alignments for evolutionary studies.
- (b) Suggest how amino acid sequences similarity may provide clues about protein functions. [5]
- Protein functions are determined by the primary structure;
 - Defined by type, number and sequence of amino acids of the polypeptide;
 - Exposed R groups interact with one another causing folding/ coiling;
 - Specific R group interactions result in specific 3D shape;
 - Resulting in protein domains with specialized functions;
 - Therefore any two polypeptides that share similarities in the amino acid sequences are likely to possess similar domains.
- (c) Describe how sequence homology searches can be useful in identifying close relatives [3]
- Involves a group of evolutionarily related proteins descended from a common ancestor;
 - Accumulation of changes to DNA sequences with time;
 - The greater the homology the closer the organisms.
- (d) Identify the benefits of using two sequence comparison softwares simultaneously. [2]
- Multiple alignment softwares such as CLUSTALW enables the construction of phylogenetic trees for evolutionary studies/ used to identify conserved regions across a group of sequences that are evolutionarily related.
 - Pairwise alignment softwares such as BLAST used to find the best-matching piecewise (local) or global alignments of two query sequences in faster.